Please write clearly in block capitals.

Centre number MFS01 | | | | | Candidate number 2026 | | | |

Surname MASTER

Forename(s) FUNCTIONAL SKILLS

Candidate signature Master Functional Skills Date: 20/10/2026

Functional Skills Level 1
MATHEMATICS
Section A: Non-Calculator Paper
Section B: Calculator Paper

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

· Complete Stepped Working & Mark Codes

· Exam Board: Open Awards

Verified by Tutor Abdul · MFS

Paper A/B

Time allowed: 30 minutes / 1 hour 30 minutes

Materials

For this paper you must have:

- mathematical instruments.

You **must not** use a calculator for section A.
You may use a calculator for section B.

Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer all questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information

- The marks for questions are shown in brackets.
- The maximum mark for section A is 15 and section B is 45.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

Advice

- In all calculations, show clearly how you work out your answer.

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SECTION A : Non-Calculator

Answer ALL questions.
Write your answers in the spaces provided.

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1 Work out $19^2 - 3 \times 5 + 2$

[2 marks]

Apply BIDMAS order of operations (Indices, Multiply, Subtract, Add):

$19^2 = 361$ --> $3 \times 5 = 15$ --> $361 - 15 = 346$ --> $346 + 2 = 348$ [M1 A1]

348 ✓

Answer

2 Calculate $17.89 + 23.47$

$17.89 + 23.47 = 41.36$

Add decimals aligning decimal points:

[1 mark]

41.36 ✓

[1 mark]

Divide by 100 (shift decimal point 2 places left):

$2.31 \div 100 = 0.0231$ [1 mark]

Answer 0.0231 ✓

3 Calculate $23.1 \div 1000$

[1 mark]

Answer

Turn over for next question

4

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4 Jeffrey records the ages of some people that attend his gym.

26	15	28	18	19	43	53	61	48
28	34	42	23	25	22	31	20	19
23	31	43	50	23	34	27	39	16

Fill in this table for Jeffrey's data.

Age	Tally	Frequency
0-19		
20-29		
30-39		
40-49	III	3
50 or over	IIII I	6
	IIII	5
	IIII	4
	II	2

Tally the 20 ages into the frequency table.

[2 marks]

Question continues on next page

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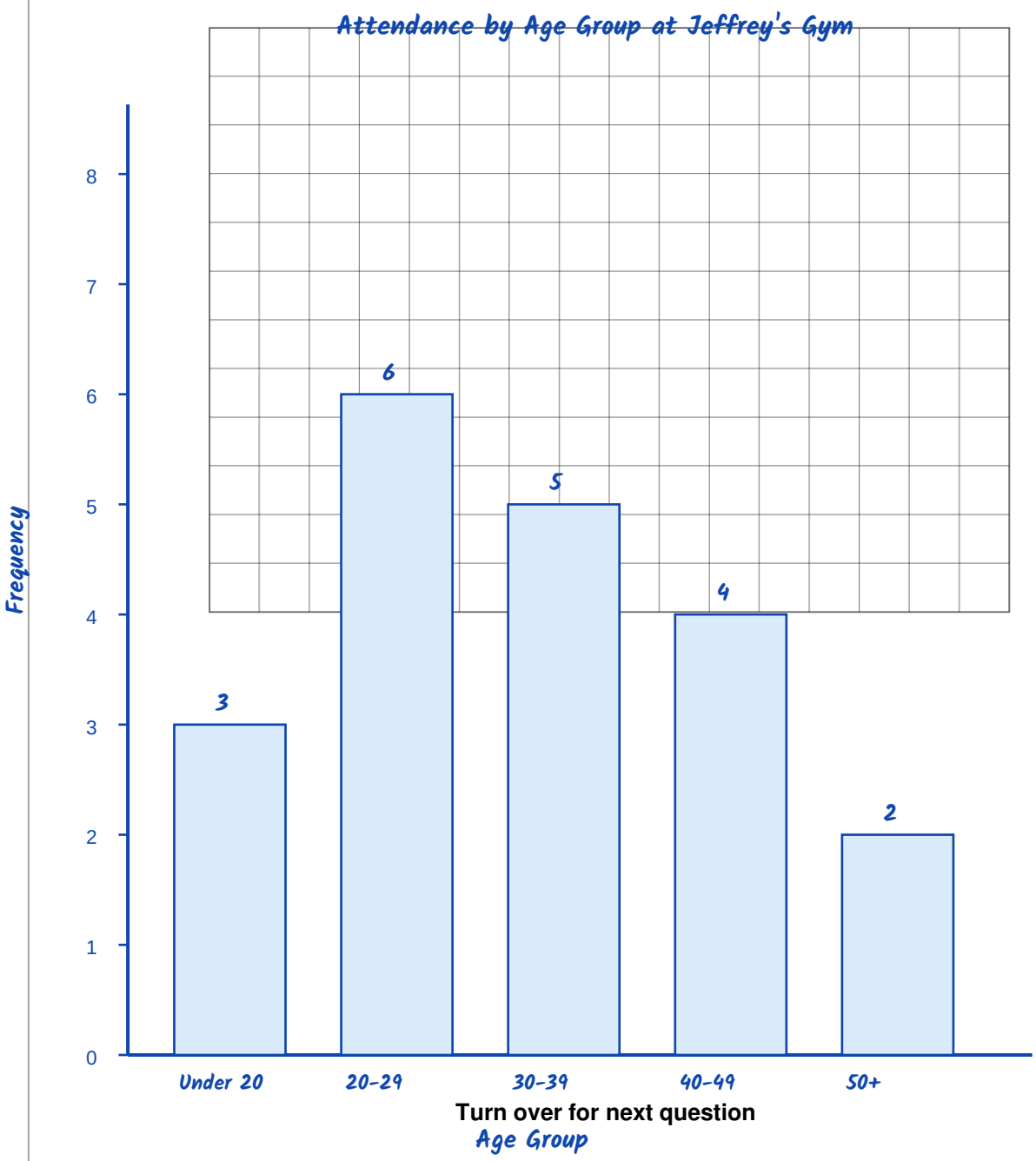
4 cont. Use this table to plot a bar chart for this data on the graph paper below.

Bar Chart: Age groups of gym attendees (Title, labelled axes, and heights):

[3 marks]



[5 marks]

Turn over for next question

5

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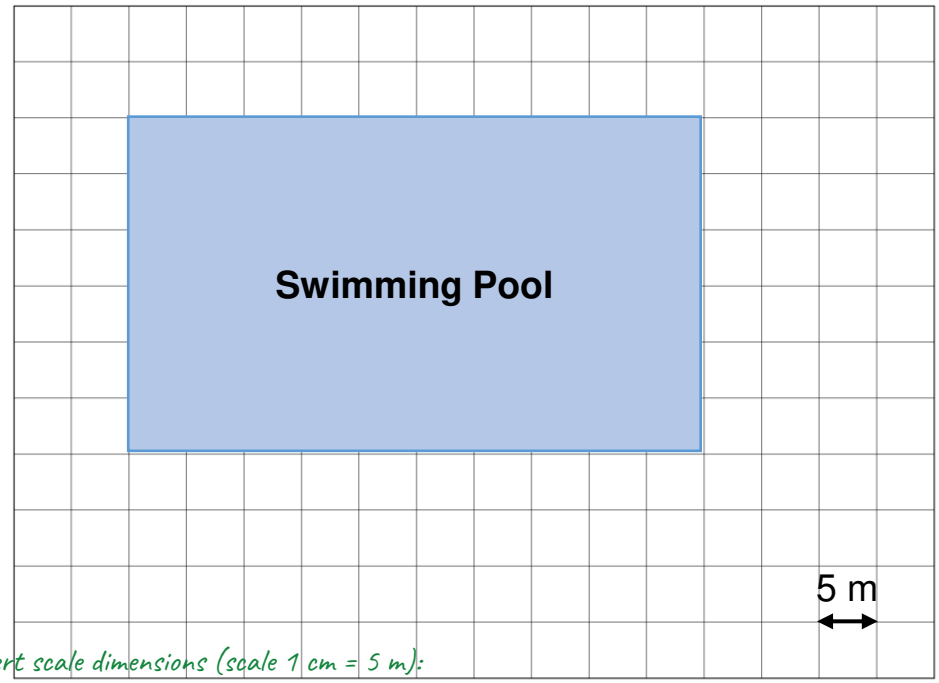
5 Jeffrey's gym has a swimming pool. Jeffrey draws a scale drawing of the swimming pool on the square grid below.

What is the actual area of the swimming pool?



Step 1: Convert scale dimensions (scale 1 cm = 5 m):

- **Length = 10 cm × 5 m = 50 metres** [M1]
- **Width = 6 cm × 5 m = 30 metres**

[3 marks]

Step 2: Area of rectangular pool = Length × Width:

Area = 50 m × 30 m = 1500 m² [M1 A1]

1500 m² ✓

Answer _____

Turn over for next question

3

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6 There are 120 members that attend Jeffrey's gym in total. The gym offer either a silver membership or a gold membership. The ratio of people on the silver membership to gold membership is 3 : 5.

How many people are there on each membership?

[2 marks]

Step 1: Divide 120 members by total ratio parts (3 + 5 = 8 parts):

1 part = 120 ÷ 8 = 15 members [M1]

Silver = 3 parts × 15 = 45 | Gold = 5 parts × 15 = 75 [A1]

Answer Silver: 45, Gold: 75 ✓

7 Jeffrey has a bag of various sports balls. He has 7 footballs, 3 rugby balls and 6 tennis balls.

What is the probability of Jeffrey picking a football out of the bag at random?

[1 mark]

Count shaded squares: 7 shaded out of 16 equal squares total:

Answer

Fraction shaded = 7 / 16 (in simplest form) [1 mark]

7 / 16 ✓

End of Section A

3

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SECTION B : Calculator

Answer ALL questions.
Write your answers in the spaces provided.

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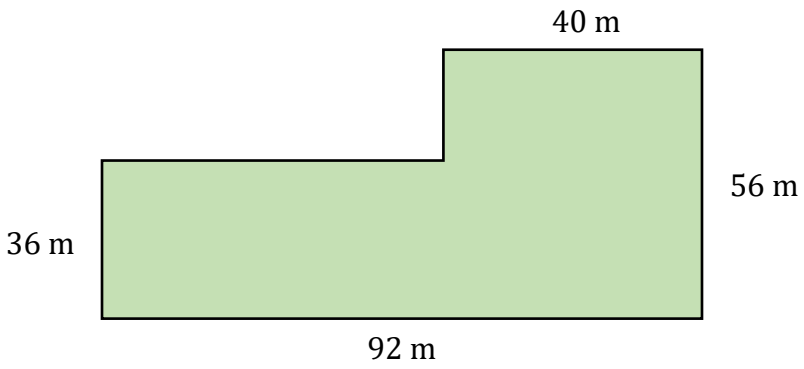
8 David is training for a charity run. As part of David's training, he runs around a field seen below. He aims to run at least 5 km.



How many **full** laps of the field will he need to run?

Assume that he can only complete **full** laps.

[3 marks]

Step 1: Calculate perimeter of the field (92m × 56m):

Perimeter = 92 + 92 + 56 + 56 = 296 metres = 0.296 km [M1]

Step 2: Number of laps to complete at least 5 km (5000m):

5 km ÷ 0.296 km = 16.891... laps [M1]

Answer

Must complete full laps: 16 laps = 4736m (< 5km) --> 17 full laps [A1]

17 full laps

Turn over for next question

3

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9 David records how long it takes him to run 5 km on 5 days in a week. He does this for 3 weeks.

He records his times in the table below.

	Time (minutes)				
	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	29	32	30	33	32
Week 2	27	29	34	30	28
Week 3	28	30	31	30	33

Calculate the range (Fastest vs Slowest) for each week:

- **Week 1: 33 − 29 = 4 mins**

[M1 A1]

In which week were the range of his times the greatest?

Show your working.

- **Week 2: 34 − 27 = 7 mins (Greatest range / most variation)**
- **Week 3: 33 − 28 = 5 mins**

[2 marks]

Week 2 (range 7 minutes) ✓

Answer _____

Calculate total cost after 20% discount on £65:

10 David buys a new pair of running shoes to use for his charity run. The price of the pair of running shoes is £65, but David has a 20% off voucher.

[1 mark]

How much will the running shoes cost David?

£65 − (0.20 × £65) = £65 − £13 = £52.00

[1 mark]

£52 ✓

Answer _____

Turn over for next question

3

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11 There are 24 members of a church. Two thirds of the members are women.

How many of the church members are women?

[1 mark]

Find two thirds (2/3) of 24 members:

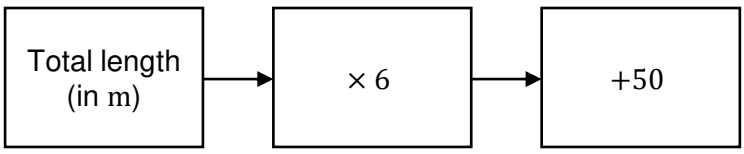
24 ÷ 3 = 8 --> 8 x 2 = 16 women [1 mark]

Answer

16 women ✓

12 Delia wants to decorate her square house by putting Christmas lights on two of its sides. Her house is 17 m wide.

To calculate the number of lights she needs, she uses this rule:



Lights come in boxes of 80.

Each box costs £15.99

Calculate how much it would cost Delia to decorate her house in this way.

[4 marks]

Step 1: Length needed = 17m x 2 sides = 34 metres

Step 2: Function machine: (34 x 6) + 50 = 204 + 50 = 254 lights [M1]

Step 3: Boxes of 80 lights needed: 254 ÷ 80 = 3.175 --> 4 boxes

Step 4: Total cost = 4 boxes x £15.99 = £63.96 [M1 A1]

£63.96 (4 boxes) ✓

Answer

Turn over for next question

5

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13 Shauna is baking cookies for 48 people. She uses the following recipe:

Cookie recipe (serves 6)
225 g butter
125 g caster sugar
300 g plain flour
100 g milk chocolate chips

Shauna has $1\frac{3}{5}$ kg of butter in her fridge.

How much more butter does she need to be able to bake the cookies for 48 people?

Include the units in your answer.

[4 marks]

Step 1: Calculate recipe scale factor (serves 6, making for 48):

Scale factor = 48 ÷ 6 = 8 [M1]

Step 2: Butter needed = 8 × 225g = 1800 grams

Butter available = 1 3/5 kg = 1.6 kg = 1600 grams [M1]

Step 3: Difference in butter = 1800g - 1600g = 200g (or 0.2 kg) [M1 A1]

Answer **200 g (0.2 kg)** ✓

Turn over for next question

4

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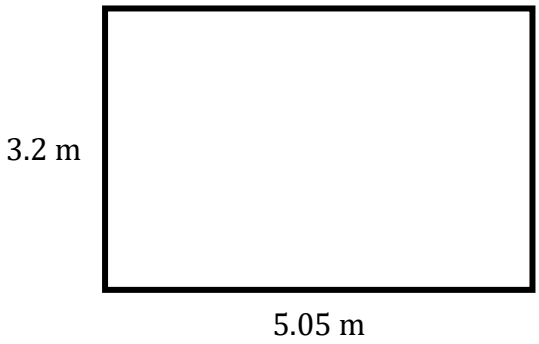
14 Bob wants to varnish his kitchen floor.

The diagram below shows the dimensions of Bob's kitchen floor.



Bob is going to give the kitchen floor 2 coats of varnish.
1 litre of varnish that Bob buys will cover 5.6 m^2 of floor.
The varnish comes in 2 litre tins, costing £14.95 per tin.

Calculate how much it would cost Bob to varnish his floor.

[5 marks]

Step 1: Area of kitchen floor = 3.2m × 5.05m = 16.16 m^2

Step 2: 2 coats of varnish = 16.16 m^2 × 2 = 32.32 m^2 **[M1]**

Step 3: Litres needed (1 litre covers 5.6 m^2):

32.32 ÷ 5.6 = 5.771... litres **[M1]**

Step 4: Tins needed (2 litre tins): 5.771 ÷ 2 = 2.885 --> 3 tins needed

Total cost = 3 tins × £14.95 = £44.85 **[M1 A1]**

Answer **£44.85 (3 tins)** ✓

Turn over for next question

5

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15 The table below shows figures for the number of ticket sales at a museum on one day.

Ticket type	Adults	Children	Concessions	Other
Tickets sold	231	105	56	12

The concession ticket type gives access to the museum to seniors and students. The ratio of seniors to students who visited the museum on this day was $4:3$.

What percentage of the total ticket sales on this day were seniors?

[4 marks]

Step 1: Ratio seniors : students = 4 : 3 for 56 concession tickets:

Total parts = 4 + 3 = 7 parts --> 1 part = 56 ÷ 7 = 8 tickets

Seniors tickets = 4 parts x 8 = 32 seniors **[M1]**

Step 2: Total tickets = 231 + 105 + 56 + 12 = 404 tickets **[M1]**

Step 3: Percentage seniors = (32 ÷ 404) x 100 = 7.9207...% --> 7.92% **[M1 A1]**

Answer **7.92%** ✓

Turn over for next question

4

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16 Billy weighs 76 kg.

Billy claims that he weighs more than the mean weight of his friends.

The weight of his friends can be seen in the table below.

Friend	Weight (kg)
Kyle	75 kg
Chloe	70 kg
Katie	68 kg
Ari	80 kg
Sean	78 kg
Mark	76 kg

Is Billy correct?

Show your working.

[3 marks]

Step 1: Sum the weights of Billy's 6 friends:

Sum = 75 + 70 + 68 + 80 + 78 + 76 = 447 kg [M1]

Step 2: Calculate mean weight = 447 ÷ 6 = 74.5 kg [M1]

Billy weighs 76 kg. 76 kg > 74.5 kg, so Yes, Billy is correct. [A1]

Answer Yes, Billy is correct (76 kg > 74.5

Turn over for next question

3

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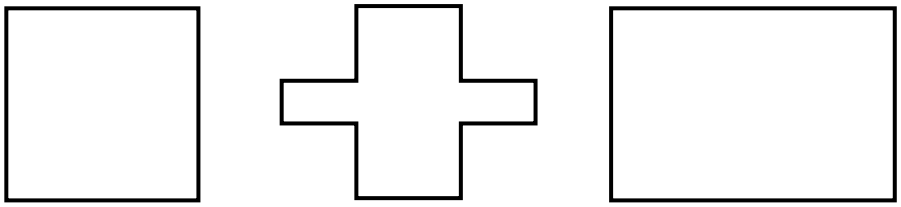
17 Which of these shapes have exactly 2 lines of symmetry?



A B C

Lines of symmetry: A (Trapezium) = 0, B (Rectangle) = 2, C (Rhombus) = 2 *[2 marks]* **[2 marks]**

Answer **B and C** ✓

18 Monique eats $\frac{1}{4}$ of a pizza and Ahmed eats 28% of the same pizza.

Who eats the most pizza?

Show your working.

[2 marks]

Convert 1/4 to a percentage: 1/4 = 25%

[M1 A1]

Monique eats 25%, Ahmed eats 28%. 28% > 25%, so Ahmed eats the most.

Answer **Ahmed (28% > 25%)** ✓

Turn over for next question

4

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19 Some Army cadets are split up into 3 teams.

The table below shows these results.

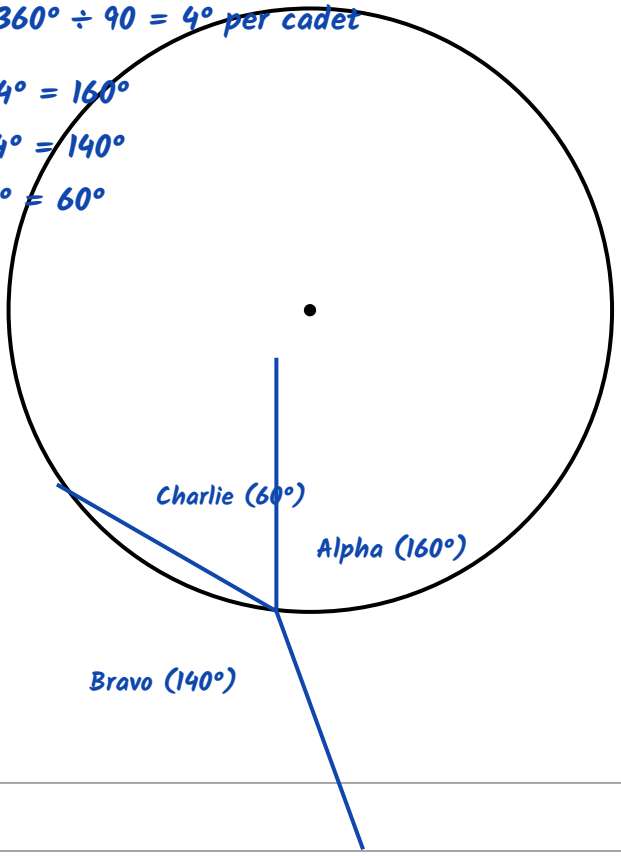
Team	Cadets
Alpha	40
Bravo	35
Charlie	15

Complete the pie chart to display this data.

Step 1: Calculate total cadets = 40 + 35 + 15 = 90 cadets

Degrees per cadet = 360° ÷ 90 = 4° per cadet [M1]

- Team Alpha : 40 × 4° = 160°
- Team Bravo : 35 × 4° = 140°
- Team Charlie: 15 × 4° = 60° [M1 A1]



[3 marks]

Answer _____

Turn over for next question

3

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20 Mary uses the following mileage chart to see the distances between various cities.

Leicester			
52	**Lincoln**		
26	36	**Nottingham**	
61	47	37	**Sheffield**

Mary wants to drive from Sheffield to Nottingham, and then from Nottingham to Leicester. She has enough petrol in her car to travel 70 miles.

Does she have enough petrol to do these journeys?

Show your working.

[2 marks]

Step 1: Read distances from the mileage chart:

- Sheffield to Nottingham = 37 miles
- Nottingham to Leicester = 26 miles

Total distance = 37 + 26 = 63 miles [M1]

Mary has petrol for 70 miles. 63 miles < 70 miles, so Yes. [A1]

Answer Yes, she has enough petrol (63 miles < 70 m

Turn over for next question

2

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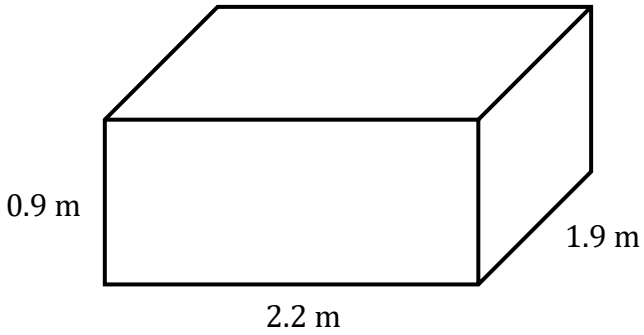
21 Wendy has bought a hot tub for her garden and wants to fill it with water.

A diagram of the hot tub can be seen below.



She is going to use a hose pipe that will fill the hot tub with $0.026\ \text{m}^3$ of water per minute.

Calculate how long it will take Wendy to fill the hot tub using this hose pipe.

Give your answer to the nearest minute.

[4 marks]

Step 1: Calculate volume of cuboid hot tub (2.2m × 1.9m × 0.9m):

Volume = 2.2 × 1.9 × 0.9 = 3.762 m³ [M1 A1]

Step 2: Hose fills at 0.026 m³ per minute. Find time:

Time = 3.762 ÷ 0.026 = 144.692... minutes [M1]

Rounding to nearest minute = 145 minutes (2 hours 25 minutes) [A1]

Answer **145 minutes** ✓

Turn over for next question

4

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22 Denise works at a Café and works as a cleaner. Her current working hours are listed in the table below.

Day	Job	Shift times	Breaks
Monday	Café	07:30 – 13:00	30 mins
Tuesday	Café	09:00 – 15:00	30 mins
Wednesday	Day Off		
Thursday	Cleaner	10:00 – 16:00	45 mins
Friday	Café	07:00 – 14:00	45 mins
Saturday	Day Off		
Sunday	Cleaner	11:30 – 14:30	-

Denise is not paid for her break times and she earns £9.10 per hour working at the Café. She earns 20% more per hour working as a cleaner.

If Denise works as a cleaner on a weekend, she gets paid an additional £20.

Work out how much Denise earns in a week.

[5 marks]

Step 1: Calculate Café hours and earnings (@ £9.10/hr):

- Mon: 5.5h - 30m = 5h | Tue: 6h - 30m = 5.5h | Fri: 7h - 45m = 6.25h

Total café hours = 16.75 hours --> 16.75 × £9.10 = £152.43 [M1]

Step 2: Calculate Cleaner hours and earnings:

- Cleaner wage: £9.10 × 1.20 = £10.92 per hour
- Thu: 6h - 45m = 5.25h | Sun: 3.0h (no break) = 3h --> 8.25 hours [M1]

Cleaner pay: (8.25 × £10.92) + £20 weekend bonus = £90.09 + £20 = £110.09

Step 3: Total weekly earnings = £152.43 + £110.09 = £262.52 [M1 A1]

Answer £262.52 ✓

5

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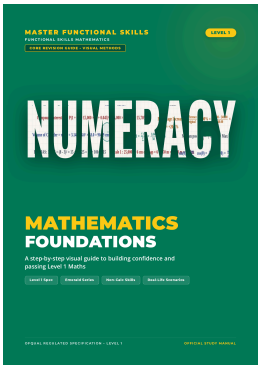

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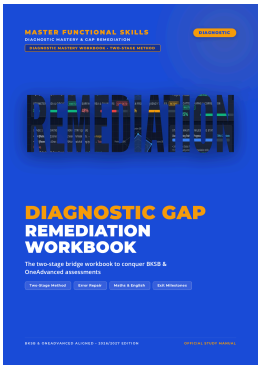

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